PA3 REPORT!!

1.How to execute your program?

在 Visual Studio Code 點擊  執行程式

(1) 首先跳出

```
----Welcome to big number calculator----
Please choose the function
1. Addition
2. Subtraction
3. Multiplication
0. Exit
-------------------------
```

並請使用者輸入要使用的函數

1. 加法

```
1
----Now for Addition----
Please enter the two numbers.
1
1
Result: 2
Mean: 1
```

2. 減法

```
2
----Now for Subtraction----
Please enter the two numbers.
1
1
Result: 0
Mean: 1
```

3. 乘法

```
3
----Now for Multiplication----
Please enter the two numbers.
5
0
Result: 0
Mean: 2
```

2. How to compile the program

(1)inputvectorandadder function

```
void inputvectorandadder(string b, string c) {
    int lengthb = b.length();
    int lengthc = c.length();
    int lengthsss;
    int tf = 1;
    if (lengthb > lengthc) {
        lengthsss = lengthb + 1;
    }
    else {
        lengthsss = lengthc + 1;
    }
    vector <int> s1(lengthsss);
    vector <int> s2(lengthsss);
    vector <int> s3(lengthsss);
    for (int i = 0;i <= lengthb - 1; ++i) {
        s1[i] = b[lengthb - 1 - i] - '0';
    }
    for (int i = 0;i <= lengthc - 1; ++i) {
        s2[i] = c[lengthc - 1 - i] - '0';
    }
    for (int z = 0; z < lengthsss; ++z) {
        if (s1[z] + s2[z] < 10) {
            s3[z] = s1[z] + s2[z];
        }
        else {
            s3[z] = s1[z] + s2[z] - 10;
            s1[z + 1]++;
        }
    }
    cout << "Result: ";
    for (int k = lengthsss - 1;k >= 0;k--) {
        if (tf == 1) {
            if (s3[k] > 0) {
                cout << s3[k];
                tf = 0;
            }
            if (s3[k] == 0) {
                tf = 1;
            }
            continue;
        }
        cout << s3[k];
    }
    if (b.length() == 1 && c.length() == 1 && s1[0] == 0 && s2[0] == 0) {
        cout << 0;
    }
```

將大數放入 vector array 裡，創一個新的 array，一位一位從後面加，當值大於 10 前一位加 1 在將值存入新的 array，最後再輸出。

(2)mean function

```
void mean(string b, string c) {
    int lengthb = b.length();
    int lengthc = c.length();
    int lengthsss;
    int tf = 1;
    if (lengthb > lengthc) {
        lengthsss = lengthb + 1;
    }
    else {
        lengthsss = lengthc + 1;
    }
    vector <int> s1(lengthsss);
    vector <int> s2(lengthsss);
    vector <int> s3(lengthsss);
    vector <int> s4(lengthsss);
    for (int i = 0;i <= lengthb - 1; ++i) {
        s1[i] = b[lengthb - 1 - i] - '0';
    }
    for (int i = 0;i <= lengthc - 1; ++i) {
        s2[i] = c[lengthc - 1 - i] - '0';
    }
    for (int z = 0; z < lengthsss; ++z) {
        if (s1[z] + s2[z] < 10) {
            s3[z] = s1[z] + s2[z];
        }
        else {
            s3[z] = s1[z] + s2[z] - 10;
            s1[z + 1]++;
        }
    }
    for (int z = lengthsss - 1;z > 0;z--) {
        if (s3[z] % 2 == 0) {
            s4[z] = s3[z] / 2;
        }
        if (s3[z] % 2 != 0) {
            s4[z] = s3[z] / 2;
            s3[z - 1] += 10;
        }
    }
    s4[0] = s3[0] / 2;
    for (int k = lengthsss - 1;k >= 0;k--) {
        if (tf == 1) {
            if (s4[k] == 0) {
                tf = 1;
            }
            if (s4[k] > 0) {
                cout << s4[k];
                tf = 0;
            }
            continue;
        }
        cout << s4[k];
    }
    if (b.length() == 1 && c.length() == 1 && s4[0] == 0) {
        cout << 0;
    }
```

Mean function 就是 adder 除以 2，所以我將 adder 複製過來後再用一個新的 array 存除以 2 的值，不過從頭開始做，當遇到奇數時下一位+10。

(1)inputvectorandSubtraction function

```
void inputvectorandSubtraction(string b, string c) {
    int lengthb = b.length();
    int lengthc = c.length();
    int lengthsss;
    int tf = 1;
    string memory;
    if (lengthb > lengthc) {
        lengthsss = lengthb + 1;
    }
    else {
        lengthsss = lengthc + 1;
    }
    vector <int> s1(lengthsss);
    vector <int> s2(lengthsss);
    vector <int> s3(lengthsss);
    for (int i = 0;i <= lengthb - 1; ++i) {
        s1[i] = b[lengthb - 1 - i] - '0';
    }
    for (int i = 0;i <= lengthc - 1; ++i) {
        s2[i] = c[lengthc - 1 - i] - '0';
    }
    for (int z = 0; z < lengthsss; ++z) {
        if (s1[z] - s2[z] >= 0) {
            s3[z] = s1[z] - s2[z];
        }
        else {
            s3[z] = s1[z] - s2[z] + 10;
            s1[z + 1]--;
        }
    }
    cout << "Result: ";
    for (int k = lengthsss - 1;k > 0;k--) {
        if (tf == 1) {
            if (s3[k] > 0) {
                cout << s3[k];
                tf = 0;
            }
            if (s3[k] == 0) {
                tf = 1;
            }
            continue;
        }
        cout << s3[k];
    }
    cout << s3[0];
}
```

也和 adder 差不多，也是從後面開始，只是變成減的，不夠的數再跟前面一位借。

(1)inputvectorMultipliaction function

```
void inputvectorMultiplication(string b, string c) {
    int lengthb = b.length();
    int lengthc = c.length();
    int tf = 1;
    int lengthsss = lengthb + lengthc;
    vector <int> s1(lengthsss);
    vector <int> s2(lengthsss);
    vector <int> s3(lengthsss);
    for (int i = 0;i <= lengthb - 1; ++i) {
        s1[i] = b[lengthb - 1 - i] - '0';
    }
    for (int i = 0;i <= lengthc - 1; ++i) {
        s2[i] = c[lengthc - 1 - i] - '0';
    }
    for (int i = 0;i < lengthc;++i) {
        int k = i;
        for (int j = 0; j < lengthb;++j) {
            s3[k] += s2[i] * s1[j];
            k++;
        }
    }
    for (int i = 0; i < lengthsss; ++i) {
        if (s3[i] > 9) {
            s3[i + 1] += s3[i] / 10;
            s3[i] = s3[i] % 10;
        }
    }
    cout << "Result: ";
    for (int k = lengthsss - 1;k >= 0;k--) {
        if (tf == 1) {
            if (s3[k] > 0) {
                cout << s3[k];
                tf = 0;
            }
            if (s3[k] == 0) {
                tf = 1;
            }
            continue;
        }
        cout << s3[k];
    }
    if ((b.length() == 1 || c.length() == 1) && (s1[0] == 0 || s2[0] == 0)) {
        cout << 0;
    }
}
```

創一個新的 array，用 2 個 for 迴圈來做，從最後一位開始累加，

在一起進位。

3.Hardness of this assignment how you overcome it

開始進入 vector 變得有點酷，但是工具也因此變多，難度適中，對於初學者的我依舊能輕鬆應付。

4.Bonus function(s) you implement if any

我加了除法呦! 還有防呆!

5. The completion of the assignment

我覺得應該是全部啦(?

6. Any suggestions about this programming assignment?

我需要一些酷酷的分數

